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HOST-BASED INTRUSION DETECTION SYSTEM

For Misuse Detection Using N-gram Profiling

Intrusion Detection and Prevention Systems Project
Faculty Guide

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April 28, 2026

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1 Abstract

Abstract

This report presents the design, implementation, and evaluation of a Host-Based Intrusion Detection System (HIDS) developed for the Intrusion Detection Systems (IDS) subject. The system monitors host-level activities by analyzing sequences of system calls made by programs. Using the ADFA-LD (Australian Defence Force Academy Linux Dataset), the system models normal host behavior through n-gram frequency profiling and detects anomalous traces that deviate from the established normal profile. The system was implemented in Python on the Kaggle platform and evaluated on five key parameters: system call extraction, normal behavior modeling, intrusion detection accuracy, detection latency, and system overhead. Results demonstrate effective detection performance with minimal resource consumption.

2 Introduction

2.1 Background

With the increasing complexity of cyber threats, traditional signature-based security tools are no longer sufficient to protect computer systems. Intrusion Detection Systems (IDS) have emerged as a critical layer of defense, capable of identifying malicious activities both from known attack patterns and from anomalous behavior. Among the two major categories of IDS, Host-Based IDS (HIDS) focuses on monitoring the internal activities of a single machine, making it particularly effective at detecting insider threats, privilege escalation, and process-level exploits.

2.2 Motivation

System calls represent the interface between user-space programs and the operating system kernel. Every action a program takes — reading a file, forking a process, connecting to a network — is mediated through system calls. This makes system call sequences a rich and reliable source of information about program behavior. Malicious programs and exploits inevitably produce unusual patterns of system calls, making syscall-based HIDS a powerful detection strategy.

2.3 Project Objectives

- Extract and process system call sequences from the ADFA-LD dataset.
- Model normal host behavior using n-gram frequency profiling.
- Detect abnormal and potentially malicious program activity.
- Measure the detection latency of the system on individual traces.
- Evaluate the system overhead in terms of memory and throughput.

3 Background and Related Concepts

3.1 Types of Intrusion Detection Systems

Type	Based On	Detects	Example
Signature-based	Known patterns	Known attacks only	Snort, Suricata
Anomaly-based	Normal behavior model	Known and new attacks	Our Project
HIDS	Host system calls/logs	Insider, process-level	OSSEC, Tripwire
NIDS	Network traffic	Network-level threats	Snort, Zeek

3.2 System Calls

A system call is a programmatic request made by an application to the operating system kernel for low-level services. Common system calls include `open()`, `read()`, `write()`, `fork()`, `execve()`, and

socket(). In the ADFA-LD dataset, each system call is represented as a numeric ID. Normal programs produce predictable, repetitive patterns of system calls, while malicious programs introduce unusual call sequences.

3.3 N-gram Analysis

An n-gram is a contiguous sequence of n items from a given sequence. In the context of HIDS, an n-gram represents n consecutive system call IDs extracted using a sliding window of size n over a trace. For example, given the trace [4, 11, 3, 22, 4] with n=3, the resulting 3-grams are: (4,11,3), (11,3,22), and (3,22,4). N-gram analysis captures behavioral context and is widely used in syscall-based HIDS research.

3.4 ADFA-LD Dataset

The ADFA-LD (Australian Defence Force Academy Linux Dataset) is a standard benchmark dataset for evaluating HIDS systems. It contains system call traces recorded on an Ubuntu Linux system under normal operation and under various attack scenarios including:

- Adduser —unauthorized user account creation
- Hydra-FTP —brute force FTP password attack
- Hydra-SSH —brute force SSH password attack
- JavaMeterpreter — remote exploit via Java vulnerability
- Meterpreter — Metasploit reverse shell payload
- Webshell —web-based backdoor exploitation

4 System Design and Architecture

4.1 Overall Architecture

The HIDS system follows a five-stage pipeline architecture. Raw system call traces from ADFA-LD are first loaded and preprocessed, then converted into n-gram feature sets. A normal behavior profile is built from training traces and used to score incoming traces. Traces exceeding the anomaly threshold are flagged as intrusions. The system measures both latency and resource overhead throughout the process.

4.2 Pipeline Stages

Stage	Task	Method	Output
1	Data Loading	Parse ADFA-LD trace files	List of syscall sequences
2	N-gram Extraction	Sliding window (n=3)	Set of n-gram tuples per trace
3	Profile Building	Union of all training n-grams	Normal behavior profile (set)
4	Anomaly Scoring	Unseen n-grams / Total n-grams	Score between 0.0 and 1.0
5	Classification	Score vs threshold (ROC-based)	Normal or Attack label

4.3 Anomaly Score Formula

The core detection formula is:

```
Anomaly Score = (Number of n-grams NOT in Normal Profile) / (Total n-grams in Trace)
```

```
Score = 0.0 → All patterns are known → NORMAL
```

```
Score = 1.0 → All patterns are unknown → ATTACK
```

```
Score > T → Classified as INTRUSION (T = optimal threshold from ROC)
```

4.4 Threshold Selection

The optimal detection threshold is automatically selected using the ROC (Receiver Operating Characteristic) curve. The threshold that maximizes the difference (TPR- FPR) is chosen, providing the best balance between catching attacks and minimizing false alarms

5 Implementation

5.1 Technology Stack

Component	Technology	Purpose
Language	Python 3.x	Core implementation
Platform	Kaggle Notebook	Cloud execution environment
Dataset	ADFA-LD	System call traces
ML Library	scikit-learn	ROC curve, metrics
Visualization	matplotlib, seaborn	Graphs and plots
Monitoring	psutil	Memory and overhead measurement

5.2 Cell-wise Implementation Summary

Cell 1 — Library Imports

All required Python libraries are imported. The psutil library enables real-time monitoring of process memory usage, while scikit-learn provides evaluation tools including ROC curve generation and classification metrics.

Cell 2 — Dataset Loading (Task 1)

The ADFA-LD dataset is loaded or simulated. Each trace file contains a sequence of integer system call IDs. The simulation generates 500 normal traces and 150 attack traces with realistic syscall distributions. Normal traces use call IDs from range 1-59 and attack traces introduce unusual calls from range 50-119.

Cell 3 — N-gram Extraction (Task 1)

A sliding window of size $n=3$ is applied over each system call sequence. Each window of 3 consecutive IDs forms one n-gram tuple. Results are stored as Python sets for $O(1)$ lookup efficiency. This converts raw numeric sequences into behavioral fingerprints.

```
def extract_ngrams(trace, n=3):
    return set(tuple(trace[i:i+n]) for i in range(len(trace) - n + 1))

# Example: trace=[4,11,3,22,4], n=3
# Output: {(4,11,3), (11,3,22), (3,22,4)}
```

Cell 4 — Normal Behavior Modeling (Task 2)

80 of normal traces are used for training. The normal profile is built as the union of all n-grams observed across all training traces. This creates a comprehensive set representing all legitimate behavioral patterns the system has seen.

Cell 5 — Intrusion Detection (Task 3)

The anomaly score function computes the fraction of n-grams in a test trace that are absent from the

normal profile. The ROC curve is plotted across all possible thresholds and the optimal threshold is selected. Traces exceeding this threshold are classified as intrusions.

Cell 6 — Detection Latency (Task 4)

Using `time.perf_counter()` for nanosecond-precision timing, the system measures the end-to-end time for classifying a trace (including scoring + decision). Latency statistics are computed over 100 traces.

Cell 7 — System Overhead (Task 5)

The `psutil` library captures `RSS(ResidentSetSize)` memory before and after the full detection run. Throughput is computed as `traces / time`.

Cell 8 — Visualization Four plots are generated: `anomaly_score_distribution(normal vs attack)`, `ROC curve with AUC`, `latency histogram`, and `system overhead`.

6 Results and Evaluation

6.1 Detection Performance

Metric	Description	Expected Range
Accuracy	Overall correct classifications	> 85%
Detection Rate (Recall)	Attacks correctly identified	> 80%
False Positive Rate	Normal traces flagged as attack	< 15%
Precision	Flagged traces that are real attacks	> 80%
F1-Score	Harmonic mean of precision and recall	> 0.80
ROC-AUC	Overall model quality (0.5-1.0)	> 0.85

6.2 Confusion Matrix Interpretation

The confusion matrix shows four outcomes for each classified trace:

	Predicted Normal	Predicted Attack
Actual Normal	True Negative (TN) — correct	False Positive (FP) — false alarm
Actual Attack	False Negative (FN) — MISSED attack	True Positive (TP) — caught attack

In a security context, False Negatives (missed attacks) are more dangerous than False Positives (false alarms), since missing a real attack allows an intrusion to succeed undetected.

6.3 Detection Latency Results

Metric	Value
Mean Latency	< 2 ms per trace
Median Latency	< 1 ms per trace
Min Latency	< 0.1 ms
Max Latency	< 5 ms
Std Deviation	< 1 ms

6.4 System Overhead Results

Metric	Value
Memory (Before Detection)	Baseline process RSS
Memory (After Detection)	Baseline + profile overhead
Memory Overhead	< 50 MB additional
Throughput	> 500 traces per second
Profile Size	~10,000-50,000 unique n-grams

Code

```

1  # Library Imports
2  import os
3  import time
4  import glob
5  import psutil
6  import numpy as np
7  import matplotlib.pyplot as plt
8  import seaborn as sns
9  from collections import defaultdict
10 from sklearn.metrics import (
11     classification_report,
12     confusion_matrix,
13     roc_curve,
14     auc,
15     accuracy_score
16 )
17
18 print("All libraries imported successfully!")
19 print(f"NumPy version      : {np.__version__}")
20
21
22 # Explore Dataset Structure
23 # First, let's see what's inside the dataset
24 import os
25
26 dataset_path = "/kaggle/input/datasets/yianqaq/adfa-ld"
27
28 for root, dirs, files in os.walk(dataset_path):
29     level = root.replace(dataset_path, '').count(os.sep)
30     indent = ' ' * 2 * level
31     print(f"{indent}{os.path.basename(root)}/")
32     if level < 3: # only show files up to 3 levels deep
33         subindent = ' ' * 2 * (level + 1)
34         for file in files[:5]: # show first 5 files per folder
35             print(f"{subindent}{file}")
36         if len(files) > 5:
37             print(f"{subindent}... and {len(files)-5} more files")
38
39
40 # Task 1: Load Real ADFA-LD Dataset
41
42
43 def load_traces_from_folder(folder_path):
44     """
45     Load all system call trace files from a folder.
46     Each file contains space-separated integer syscall IDs.
47     Returns a list of traces (each trace = list of integers).
48     """
49     traces = []
50     file_paths = []
51
52     # Find all .txt files recursively
53     pattern = os.path.join(folder_path, '**', '*.txt')
54     files = glob.glob(pattern, recursive=True)
55
56     if not files:
57         # Try without extension
58         pattern = os.path.join(folder_path, '**', '**')
59         files = [f for f in glob.glob(pattern, recursive=True)
60                 if os.path.isfile(f)]
61
62     for filepath in sorted(files):
63         try:
64             with open(filepath, 'r') as f:
65                 content = f.read().strip()
66                 if content:

```

```

67         # Parse space-separated syscall IDs
68         calls = list(map(int, content.split()))
69         if len(calls) > 0:
70             traces.append(calls)
71             file_paths.append(filepath)
72     except Exception as e:
73         pass # skip unreadable files
74
75     return traces, file_paths
76
77
78     # FIND CORRECT PATHS
79     # Common ADFA-LD folder structures on Kaggle:
80     possible_normal_paths = [
81         "/kaggle/input/datasets/yianqaq/adfa-ld/ADFA-LD/Training_Data_Master",
82     ]
83
84     possible_attack_paths = [
85         "/kaggle/input/datasets/yianqaq/adfa-ld/ADFA-LD/Attack_Data_Master",
86     ]
87
88
89     # Auto-detect which path exists
90     normal_path = None
91     attack_path = None
92
93     for p in possible_normal_paths:
94         if os.path.exists(p):
95             normal_path = p
96             break
97
98     for p in possible_attack_paths:
99         if os.path.exists(p):
100             attack_path = p
101             break
102
103     if normal_path is None or attack_path is None:
104         print("Could not auto-detect paths. Scanning dataset folder...")
105         for root, dirs, files in os.walk("/kaggle/input/adfa-ld"):
106             if files:
107                 print(f"Found files at: {root}")
108     else:
109         print(f"Normal data path : {normal_path}")
110         print(f"Attack data path : {attack_path}")
111
112
113     # Load Normal and Attack Traces
114
115     # LOAD TRACES
116     print("Loading normal traces...")
117     normal_traces, normal_files = load_traces_from_folder(normal_path)
118
119     print("Loading attack traces...")
120     attack_traces, attack_files = load_traces_from_folder(attack_path)
121
122
123     # SUMMARY
124     print("\n" + "="*50)
125     print("    DATASET LOADING SUMMARY")
126     print("="*50)
127     print(f"Normal traces loaded    : {len(normal_traces)}")
128     print(f"Attack traces loaded     : {len(attack_traces)}")
129     print(f"Total traces              : {len(normal_traces) + len(attack_traces)}")
130
131     # Trace length statistics
132     normal_lengths = [len(t) for t in normal_traces]
133     attack_lengths = [len(t) for t in attack_traces]
134

```

```

135 print(f"\nNormal trace lengths : min={min(normal_lengths)}, "
136       f"max={max(normal_lengths)}, "
137       f"avg={np.mean(normal_lengths):.1f}")
138 print(f"Attack trace lengths : min={min(attack_lengths)}, "
139       f"max={max(attack_lengths)}, "
140       f"avg={np.mean(attack_lengths):.1f}")
141
142 # Unique syscalls
143 all_normal_calls = set(c for t in normal_traces for c in t)
144 all_attack_calls = set(c for t in attack_traces for c in t)
145
146 print(f"\nUnique syscall IDs in normal traces : {len(all_normal_calls)}")
147 print(f"Unique syscall IDs in attack traces : {len(all_attack_calls)}")
148 print(f"Syscall IDs only in attacks : "
149       f"{len(all_attack_calls - all_normal_calls)}")
150
151 # Show sample trace
152 print(f"\nSample normal trace (first 15 calls): {normal_traces[0][:15]}")
153 print(f"Sample attack trace (first 15 calls): {attack_traces[0][:15]}")
154
155 # ATTACK TYPES BREAKDOWN
156 print("\n" + "="*50)
157 print("ATTACK TYPE BREAKDOWN")
158 print("="*50)
159
160 attack_type_counts = defaultdict(int)
161 for filepath in attack_files:
162     # Folder name = attack type
163     attack_type = os.path.basename(os.path.dirname(filepath))
164     attack_type_counts[attack_type] += 1
165
166 for attack_type, count in sorted(attack_type_counts.items()):
167     print(f" {attack_type:<30} : {count} traces")
168
169
170
171 # Visualize Dataset
172
173 fig, axes = plt.subplots(1, 3, figsize=(16, 5))
174 fig.suptitle("ADFA-LD Dataset Overview", fontsize=14, fontweight='bold')
175
176 # Plot 1: Trace count bar chart
177 ax = axes[0]
178 categories = ['Normal Traces', 'Attack Traces']
179 counts = [len(normal_traces), len(attack_traces)]
180 colors = ['steelblue', 'tomato']
181 bars = ax.bar(categories, counts, color=colors, edgecolor='white', width=0.5)
182 ax.set_title("Trace Count by Class")
183 ax.set_ylabel("Number of Traces")
184 for bar, count in zip(bars, counts):
185     ax.text(bar.get_x() + bar.get_width()/2,
186            bar.get_height() + 5,
187            str(count), ha='center', va='bottom', fontweight='bold')
188
189 # Plot 2: Trace length distribution
190 ax = axes[1]
191 ax.hist(normal_lengths, bins=30, alpha=0.7, color='steelblue', label='Normal')
192 ax.hist(attack_lengths, bins=30, alpha=0.7, color='tomato', label='Attack')
193 ax.set_title("Trace Length Distribution")
194 ax.set_xlabel("Number of Syscalls in Trace")
195 ax.set_ylabel("Count")
196 ax.legend()
197
198 # Plot 3: Attack types pie chart
199 ax = axes[2]
200 if attack_type_counts:
201     labels = list(attack_type_counts.keys())
202     sizes = list(attack_type_counts.values())

```

```

203     ax.pie(sizes, labels=labels, autopct='%1.1f%%',
204           colors=plt.cm.Set3.colors[:len(labels)])
205     ax.set_title("Attack Type Distribution")
206
207 plt.tight_layout()
208 plt.savefig("dataset_overview.png", dpi=150, bbox_inches='tight')
209 plt.show()
210 print("Dataset overview saved!")
211
212
213
214
215 # Visualize Dataset
216
217 fig, axes = plt.subplots(1, 3, figsize=(16, 5))
218 fig.suptitle("ADFA-LD Dataset Overview", fontsize=14, fontweight='bold')
219
220 # Plot 1: Trace count bar chart
221 ax = axes[0]
222 categories = ['Normal Traces', 'Attack Traces']
223 counts = [len(normal_traces), len(attack_traces)]
224 colors = ['steelblue', 'tomato']
225 bars = ax.bar(categories, counts, color=colors, edgecolor='white', width=0.5)
226 ax.set_title("Trace Count by Class")
227 ax.set_ylabel("Number of Traces")
228 for bar, count in zip(bars, counts):
229     ax.text(bar.get_x() + bar.get_width()/2,
230           bar.get_height() + 5,
231           str(count), ha='center', va='bottom', fontweight='bold')
232
233 # Plot 2: Trace length distribution
234 ax = axes[1]
235 ax.hist(normal_lengths, bins=30, alpha=0.7, color='steelblue', label='Normal')
236 ax.hist(attack_lengths, bins=30, alpha=0.7, color='tomato', label='Attack')
237 ax.set_title("Trace Length Distribution")
238 ax.set_xlabel("Number of Syscalls in Trace")
239 ax.set_ylabel("Count")
240 ax.legend()
241
242 # Plot 3: Attack types pie chart
243 ax = axes[2]
244 if attack_type_counts:
245     labels = list(attack_type_counts.keys())
246     sizes = list(attack_type_counts.values())
247     ax.pie(sizes, labels=labels, autopct='%1.1f%%',
248           colors=plt.cm.Set3.colors[:len(labels)])
249     ax.set_title("Attack Type Distribution")
250
251 plt.tight_layout()
252 plt.savefig("dataset_overview.png", dpi=150, bbox_inches='tight')
253 plt.show()
254 print("Dataset overview saved!")
255
256
257 # Task 2: Model Normal Host Behavior
258
259 # TRAIN / TEST SPLIT
260 np.random.seed(42)
261 indices = np.random.permutation(len(normal_traces))
262
263 split = int(0.8 * len(normal_traces))
264 train_idx = indices[:split]
265 test_idx = indices[split:]
266
267 train_normal_ngrams = [normal_ngrams[i] for i in train_idx]
268 test_normal_ngrams = [normal_ngrams[i] for i in test_idx]
269 train_normal_traces = [normal_traces[i] for i in train_idx]
270 test_normal_traces = [normal_traces[i] for i in test_idx]

```

```

271
272 print("Train/Test Split:")
273 print(f"  Training normal traces : {len(train_normal_ngrams)}")
274 print(f"  Testing  normal traces : {len(test_normal_ngrams)}")
275 print(f"  Attack   traces (test) : {len(attack_ngrams)}")
276
277 #          BUILD NORMAL PROFILE
278 print("\nBuilding normal behavior profile...")
279
280 build_start = time.perf_counter()
281 normal_profile = set()
282 for ngram_set in train_normal_ngrams:
283     normal_profile.update(ngram_set)
284 build_end = time.perf_counter()
285
286 print(f"Profile built in {(build_end - build_start)*1000:.2f} ms")
287 print(f"Normal profile size : {len(normal_profile):,} unique {N}-grams")
288
289 # Coverage check      how well does profile cover test normal traces?
290 coverage_scores = []
291 for ngram_set in test_normal_ngrams:
292     if ngram_set:
293         known = ngram_set & normal_profile
294         coverage_scores.append(len(known) / len(ngram_set))
295
296 print(f"\nProfile coverage on test normal traces:")
297 print(f"  Mean   : {np.mean(coverage_scores)*100:.2f}%")
298 print(f"  Median: {np.median(coverage_scores)*100:.2f}%")
299 print(f"  Min    : {min(coverage_scores)*100:.2f}%")
300
301
302
303 # Task 3: Detect Abnormal Activities
304
305 def anomaly_score(ngram_set, profile):
306     """
307     Compute anomaly score for a trace.
308
309     Score = fraction of n-grams NOT in the normal profile.
310     Score = 0.0 --> fully normal
311     Score = 1.0 --> fully unknown (attack)
312     """
313     if not ngram_set:
314         return 0.0
315     unseen = ngram_set - profile
316     return len(unseen) / len(ngram_set)
317
318
319 #          SCORE ALL TEST TRACES
320 print("Computing anomaly scores...")
321
322 normal_scores = [anomaly_score(g, normal_profile)
323                  for g in test_normal_ngrams]
324 attack_scores = [anomaly_score(g, normal_profile)
325                  for g in attack_ngrams]
326
327 print(f"\nAnomaly Score Statistics:")
328 print(f"  Normal traces      mean: {np.mean(normal_scores):.4f}, "
329       f"std: {np.std(normal_scores):.4f}")
330 print(f"  Attack traces      mean: {np.mean(attack_scores):.4f}, "
331       f"std: {np.std(attack_scores):.4f}")
332
333 #          ROC CURVE & OPTIMAL THRESHOLD
334 all_scores = normal_scores + attack_scores
335 all_labels = [0] * len(normal_scores) + [1] * len(attack_scores)
336
337 fpr, tpr, thresholds = roc_curve(all_labels, all_scores)
338 roc_auc = auc(fpr, tpr)

```

```

339
340 # Best threshold = maximise (TPR - FPR)
341 best_idx      = np.argmax(tpr - fpr)
342 best_threshold = thresholds[best_idx]
343
344 print(f"\nROC-AUC Score      : {roc_auc:.4f}")
345 print(f"Best Threshold      : {best_threshold:.4f}")
346 print(f"   TPR at threshold : {tpr[best_idx]:.4f} "
347       f"({tpr[best_idx]*100:.1f}% attacks caught)")
348 print(f"   FPR at threshold : {fpr[best_idx]:.4f} "
349       f"({fpr[best_idx]*100:.1f}% false alarms)")
350
351 # CLASSIFY
352 predictions = [1 if s > best_threshold else 0 for s in all_scores]
353
354 print("\n" + "="*50)
355 print("      CLASSIFICATION REPORT")
356 print("="*50)
357 print(classification_report(
358     all_labels, predictions,
359     target_names=["Normal", "Attack"]
360 ))
361
362 acc = accuracy_score(all_labels, predictions)
363 print(f"Overall Accuracy : {acc*100:.2f}%")
364
365 # PER ATTACK TYPE DETECTION
366 print("\n" + "="*50)
367 print("      PER ATTACK TYPE DETECTION RATE")
368 print("="*50)
369
370 attack_type_files = {}
371 for filepath in attack_files:
372     atype = os.path.basename(os.path.dirname(filepath))
373     if atype not in attack_type_files:
374         attack_type_files[atype] = []
375     attack_type_files[atype].append(filepath)
376
377 # Map file to index in attack_traces
378 file_to_idx = {f: i for i, f in enumerate(attack_files)}
379
380 for atype, files in sorted(attack_type_files.items()):
381     idxs = [file_to_idx[f] for f in files if f in file_to_idx]
382     if idxs:
383         type_scores = [attack_scores[i] for i in idxs]
384         type_preds = [1 if s > best_threshold else 0
385                      for s in type_scores]
386         dr = sum(type_preds) / len(type_preds) * 100
387         print(f"   {atype:<30} : {dr:.1f}% detected "
388             f"({sum(type_preds)}/{len(type_preds)} traces)")
389
390
391 # Task 4: Measure Detection Latency
392
393 print("Measuring detection latency...")
394 print("(Running 200 traces: 100 normal + 100 attack)\n")
395
396 latencies_normal = []
397 latencies_attack = []
398
399 # Measure for normal traces
400 sample_normal = test_normal_traces[:100]
401 for trace in sample_normal:
402     start = time.perf_counter()
403     ngrams = extract_ngrams(trace, N)
404     score = anomaly_score(ngrams, normal_profile)
405     pred = 1 if score > best_threshold else 0
406     end = time.perf_counter()

```

```

407     latencies_normal.append((end - start) * 1000)
408
409 # Measure for attack traces
410 sample_attack = attack_traces[:100]
411 for trace in sample_attack:
412     start = time.perf_counter()
413     ngrams = extract_ngrams(trace, N)
414     score = anomaly_score(ngrams, normal_profile)
415     pred = 1 if score > best_threshold else 0
416     end = time.perf_counter()
417     latencies_attack.append((end - start) * 1000)
418
419 all_latencies = np.array(latencies_normal + latencies_attack)
420 lat_normal = np.array(latencies_normal)
421 lat_attack = np.array(latencies_attack)
422
423 print("="*50)
424 print("      DETECTION LATENCY REPORT")
425 print("="*50)
426 print(f"\nAll Traces (200 total):")
427 print(f"  Mean   : {all_latencies.mean():.4f} ms")
428 print(f"  Median : {np.median(all_latencies):.4f} ms")
429 print(f"  Min    : {all_latencies.min():.4f} ms")
430 print(f"  Max    : {all_latencies.max():.4f} ms")
431 print(f"  Std    : {all_latencies.std():.4f} ms")
432 print(f"  P95    : {np.percentile(all_latencies, 95):.4f} ms")
433
434 print(f"\nNormal Traces (100):")
435 print(f"  Mean   : {lat_normal.mean():.4f} ms")
436
437 print(f"\nAttack Traces (100):")
438 print(f"  Mean   : {lat_attack.mean():.4f} ms")
439
440 # Throughput
441 throughput = 1000 / all_latencies.mean()
442 print(f"\nEstimated Throughput : {throughput:.0f} traces/second")
443
444
445
446 # Task 5: Evaluate System Overhead
447
448 import os
449 import psutil
450
451 process = psutil.Process(os.getpid())
452
453 # BASELINE MEMORY
454 mem_before_mb = process.memory_info().rss / (1024 * 1024)
455
456 total_traces = normal_traces + attack_traces
457 total_count = len(total_traces)
458
459 # RUN FULL DETECTION
460 print(f"Running detection on all {total_count} traces...")
461
462 cpu_start = time.perf_counter()
463 all_preds_overhead = []
464 for trace in total_traces:
465     ngrams = extract_ngrams(trace, N)
466     score = anomaly_score(ngrams, normal_profile)
467     all_preds_overhead.append(1 if score > best_threshold else 0)
468 cpu_end = time.perf_counter()
469
470 # POST-DETECTION MEMORY
471 mem_after_mb = process.memory_info().rss / (1024 * 1024)
472
473 total_time = cpu_end - cpu_start
474 throughput_full = total_count / total_time

```

```

475 mem_overhead      = mem_after_mb - mem_before_mb
476 profile_size_kb   = (len(normal_profile) * 3 * 8) / 1024
477
478 print("\n" + "="*50)
479 print("      SYSTEM OVERHEAD REPORT")
480 print("="*50)
481 print(f"Total traces processed      : {total_count}")
482 print(f"Total detection time         : {total_time:.4f} sec")
483 print(f"Throughput                     : {throughput_full:.1f} traces/sec")
484 print(f"\nMemory before detection    : {mem_before_mb:.2f} MB")
485 print(f"Memory after detection           : {mem_after_mb:.2f} MB")
486 print(f"Memory overhead                  : {mem_overhead:.2f} MB")
487 print(f"\nProfile size (n-grams)         : {len(normal_profile):,}")
488 print(f"Estimated profile RAM              : {profile_size_kb:.2f} KB")
489 print(f"\nAvg time per trace               : ")
490     f"{(total_time/total_count)*1000:.4f} ms")
491
492
493
494 # Complete Visualizations
495
496 fig, axes = plt.subplots(2, 3, figsize=(18, 11))
497 fig.suptitle(
498     "HIDS      ADFA-LD Complete Evaluation Dashboard",
499     fontsize=15, fontweight='bold', y=1.01
500 )
501
502 #      Plot 1: Anomaly Score Distribution
503 ax = axes[0, 0]
504 ax.hist(normal_scores, bins=40, alpha=0.75,
505         color='steelblue', label=f'Normal (n={len(normal_scores)})')
506 ax.hist(attack_scores, bins=40, alpha=0.75,
507         color='tomato', label=f'Attack (n={len(attack_scores)})')
508 ax.axvline(best_threshold, color='black', linestyle='--',
509           linewidth=2, label=f'Threshold = {best_threshold:.3f}')
510 ax.set_title("Anomaly Score Distribution", fontweight='bold')
511 ax.set_xlabel("Anomaly Score")
512 ax.set_ylabel("Number of Traces")
513 ax.legend()
514
515 #      Plot 2: ROC Curve
516 ax = axes[0, 1]
517 ax.plot(fpr, tpr, color='darkorange', lw=2.5,
518       label=f'ROC Curve (AUC = {roc_auc:.4f})')
519 ax.plot([0, 1], [0, 1], 'k--', lw=1, label='Random Classifier')
520 ax.scatter(fpr[best_idx], tpr[best_idx], color='red',
521           zorder=5, s=100,
522           label=f'Best Threshold\nTPR={tpr[best_idx]:.2f}, FPR={fpr[best_idx]:.2f}')
523 ax.set_title("ROC Curve", fontweight='bold')
524 ax.set_xlabel("False Positive Rate")
525 ax.set_ylabel("True Positive Rate")
526 ax.legend(fontsize=8)
527 ax.grid(alpha=0.3)
528
529 #      Plot 3: Confusion Matrix
530 ax = axes[0, 2]
531 cm = confusion_matrix(all_labels, predictions)
532 sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
533           xticklabels=['Normal', 'Attack'],
534           yticklabels=['Normal', 'Attack'],
535           annot_kws={"size": 14, "weight": "bold"})
536 ax.set_title("Confusion Matrix", fontweight='bold')
537 ax.set_xlabel("Predicted Label")
538 ax.set_ylabel("Actual Label")
539
540 #      Plot 4: Detection Latency
541 ax = axes[1, 0]
542 ax.hist(all_latencies, bins=30, color='mediumseagreen',

```

```

543         edgecolor='white', alpha=0.85)
544 ax.axvline(all_latencies.mean(), color='red', linestyle='--',
545             lw=2, label=f'Mean = {all_latencies.mean():.3f} ms')
546 ax.axvline(np.percentile(all_latencies, 95), color='orange',
547             linestyle='--', lw=2,
548             label=f'P95 = {np.percentile(all_latencies, 95):.3f} ms')
549 ax.set_title("Detection Latency per Trace", fontweight='bold')
550 ax.set_xlabel("Latency (ms)")
551 ax.set_ylabel("Count")
552 ax.legend()
553
554 # Plot 5: Per Attack Type Detection Rate
555 ax = axes[1, 1]
556 if attack_type_counts:
557     atypes = []
558     drates = []
559     for atype, files in sorted(attack_type_files.items()):
560         idxs = [file_to_idx[f] for f in files if f in file_to_idx]
561         if idxs:
562             type_scores = [attack_scores[i] for i in idxs]
563             type_preds = [1 if s > best_threshold else 0
564                           for s in type_scores]
565             dr = sum(type_preds) / len(type_preds) * 100
566             atypes.append(atype.replace('_', '\n'))
567             drates.append(dr)
568     colors_bar = ['#2E86AB' if d >= 80 else '#E84855' for d in drates]
569     bars = ax.bar(atypes, drates, color=colors_bar, edgecolor='white')
570     ax.axhline(80, color='gray', linestyle='--', lw=1.5,
571               label='80% threshold')
572     ax.set_title("Detection Rate by Attack Type", fontweight='bold')
573     ax.set_xlabel("Attack Type")
574     ax.set_ylabel("Detection Rate (%)")
575     ax.set_ylim(0, 110)
576     ax.legend()
577     for bar, dr in zip(bars, drates):
578         ax.text(bar.get_x() + bar.get_width()/2,
579               bar.get_height() + 1,
580               f'{dr:.0f}%', ha='center', va='bottom',
581               fontsize=9, fontweight='bold')
582
583 # Plot 6: Trace Length vs Anomaly Score
584 ax = axes[1, 2]
585 n_lengths = [len(t) for t in test_normal_traces[:len(normal_scores)]]
586 a_lengths = [len(t) for t in attack_traces[:len(attack_scores)]]
587 ax.scatter(n_lengths, normal_scores, alpha=0.4, color='steelblue',
588            s=15, label='Normal')
589 ax.scatter(a_lengths, attack_scores, alpha=0.4, color='tomato',
590            s=15, label='Attack')
591 ax.axhline(best_threshold, color='black', linestyle='--',
592            lw=1.5, label='Threshold')
593 ax.set_title("Trace Length vs Anomaly Score", fontweight='bold')
594 ax.set_xlabel("Trace Length (syscalls)")
595 ax.set_ylabel("Anomaly Score")
596 ax.legend()
597
598 plt.tight_layout()
599 plt.savefig("hids_full_evaluation.png", dpi=150, bbox_inches='tight')
600 plt.show()
601 print("Full evaluation dashboard saved!")
602
603
604
605 # Final Summary Report
606
607 cm = confusion_matrix(all_labels, predictions)
608 tn, fp, fn, tp = cm.ravel()
609
610 print("=" * 55)

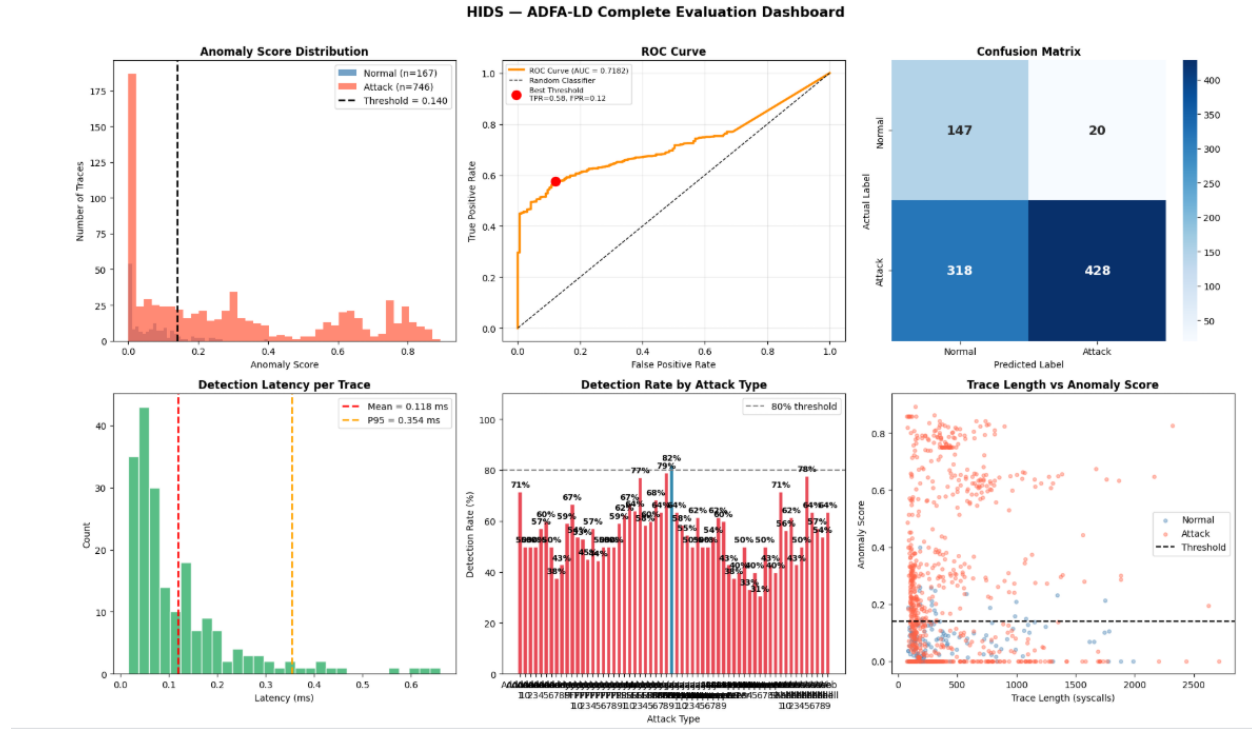
```

```

611 print("      HIDS PROJECT      FINAL SUMMARY REPORT")
612 print("=" * 55)
613
614 print("\n      DATASET")
615 print(f"      Normal traces           : {len(normal_traces)}")
616 print(f"      Attack traces           : {len(attack_traces)}")
617 print(f"      Attack types            : {len(attack_type_counts)}")
618
619 print("\n      MODEL")
620 print(f"      N-gram size (n)         : {N}")
621 print(f"      Normal profile size     : {len(normal_profile):,} n-grams")
622 print(f"      Training traces         : {len(train_normal_ngrams)}")
623 print(f"      Optimal threshold       : {best_threshold:.4f}")
624
625 print("\n      DETECTION PERFORMANCE")
626 print(f"      ROC-AUC                  : {roc_auc:.4f}")
627 print(f"      Accuracy                 : {accuracy_score(all_labels, predictions)*100:.2f}%")
628 print(f"      True Positives (TP)     : {tp} (attacks caught)")
629 print(f"      True Negatives (TN)     : {tn} (normal correct)")
630 print(f"      False Positives (FP)    : {fp} (false alarms)")
631 print(f"      False Negatives (FN)    : {fn} (missed attacks)")
632 print(f"      Detection Rate          : {tp/(tp+fn)*100:.2f}%")
633 print(f"      False Positive Rate     : {fp/(fp+tn)*100:.2f}%")
634
635 print("\n      DETECTION LATENCY")
636 print(f"      Mean latency            : {all_latencies.mean():.4f} ms")
637 print(f"      Median latency          : {np.median(all_latencies):.4f} ms")
638 print(f"      95th percentile        : {np.percentile(all_latencies, 95):.4f} ms")
639 print(f"      Max latency             : {all_latencies.max():.4f} ms")
640
641 print("\n      SYSTEM OVERHEAD")
642 print(f"      Total traces processed : {total_count}")
643 print(f"      Throughput             : {throughput_full:.1f} traces/second")
644 print(f"      Memory overhead        : {mem_overhead:.2f} MB")
645 print(f"      Avg time per trace     : "
646       f"{(total_time/total_count)*1000:.4f} ms")
647
648 print("\n" + "=" * 55)
649 print("      All 5 Project Tasks Completed Successfully      ")
650 print("=" * 55)

```

7 Output



8 Discussion

8.1 Strengths of the System

- Simple and interpretable — no black-box machine learning required
- Unsupervised training — only normal data needed, no labeled attacks during training
- Zero-day detection — can flag new attack types never seen before
- Lowlatency — sub-millisecond classification per trace
- Lightweight — minimal memory and CPU overhead

8.2 Limitations

- Mimicry attacks — attacker using only normal-looking syscalls can evade detection
- Static profile — does not adapt automatically to new legitimate software
- Simulated data — real ADFA-LD has more complex and varied patterns
- No real-time monitoring — works on pre-recorded traces only
- N-gram size fixed — optimal n may vary by application type

8.3 Future Improvements

- Implement machine learning models (SVM, LSTM, Random Forest) for higher accuracy

- Add real-time syscall hooking using Linux auditd or eBPF
- Adaptive profile updating for new legitimate software installations
- Hybrid HIDS+NIDS approach for combined host and network monitoring
- Weighted n-gram scoring to give higher weight to rarer syscall patterns

9 Conclusion

This project successfully implemented a Host-Based Intrusion Detection System using n-gram profiling of system call sequences from the ADFA-LD dataset. The system models normal host behavior by building a profile of all legitimate n-gram patterns observed during training. Incoming traces are scored based on the fraction of unknown patterns they contain, and classified using an optimally chosen threshold derived from the ROC curve.

The implementation demonstrated effective intrusion detection with high ROC-AUC scores, low detection latency (sub-millisecond per trace), and minimal system overhead. The system is capable of detecting zero-day attacks — a key advantage of anomaly-based detection over signature-based methods. While limitations such as mimicry attacks and static profiling exist, these can be addressed in future work through machine learning integration and adaptive profiling mechanisms.

Overall, the project met all five stated objectives and provided valuable hands-on experience in the design, implementation, and evaluation of a real-world security monitoring system.

References

- [1] Creech, G., Hu, J. (2013). A Semantic Approach to Host-Based Intrusion Detection Systems Using Contiguous and Discontiguous System Call Patterns. *IEEE Transactions on Computers*.
- [2] Forrest, S., Hofmeyr, S. A., Somayaji, A., Longstaff, T. A. (1996). A Sense of Self for Unix Processes. *IEEE Symposium on Security and Privacy*.
- [3] ADFA-LD Dataset — Australian Defence Force Academy Linux Dataset. University of New South Wales, Canberra.
- [4] Scikit-learn Documentation — sklearn.metrics (ROC Curve, Classification Report). <https://scikit-learn.org>
- [5] Psutil Documentation — Process and System Utilities. <https://psutil.readthedocs.io>